Multiplicity Thresholds for Sidon Divisor Sets and Cyclic Triple Collisions

Supporting note for OEIS A398433, A398434, A398435, and A398577

Felix Huber
Independent Researcher, Luzern, Switzerland
ORCID 0009-0005-1568-1579

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Abstract

This paper is the common supporting note for OEIS sequences A398433, A398434, A398435, and A398577, which describe four aspects of additive collisions in sets of positive divisors: where a collision first appears under divisibility, how large a Sidon subset can remain, the total excess of repeated representations, and the largest multiplicity of a single sum. Writing $\mathcal{D}(n)$ for the positive divisors of n and $R(n)$ for the maximum number of unordered representations of one integer as $x + y$ with $x, y \in \mathcal{D}(n)$, we prove that for squarefree divisibility-minimal non-Sidon integers

$$R(n) \geq 3 \Longrightarrow \omega(n) \geq 5, \qquad R(n) \geq 4 \Longrightarrow \omega(n) \geq 6.$$

Both bounds are sharp. The two smallest minimal examples with $R(n) = 3$ are

$$7 \cdot 11 \cdot 31 \cdot 89 \cdot 383 \quad \text{and} \quad 5 \cdot 11 \cdot 41 \cdot 67 \cdot 547.$$

A core–defect decomposition reduces the threshold questions to finite collections of prime-factor orientations, which are closed by exact sign and modular certificates together with a small number of elementary arguments. All computer-assisted steps are accompanied by machine-readable certificate ledgers and independent exact verifiers. The two smallest examples are then explained by a simple cyclic linear lemma, leading to two triple-collision templates and four explicit polynomial families. Every admissible member of these families is simultaneously a term of A398433 and satisfies A398434$(n) = 30$, A398435$(n) = 2$, and A398577$(n) = 3$; under Schinzel's Hypothesis H, each family contains infinitely many such integers.

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1 Introduction

A Sidon set is a set in which an unordered pair is determined by its sum. Thus a finite set $A \subset \mathbb{Z}$ is Sidon when

$$x + y = u + v, \qquad x \leq y, \quad u \leq v,$$

with $x, y, u, v \in A$ forces $\{x, y\} = \{u, v\}$. The classical theory asks how large such sets can be and how their representation functions behave; see, for example, [1, 3, 5].

Here the ambient set is fixed by multiplication rather than addition. For a positive integer n we write

$$\mathcal{D}(n) = \{d \in \mathbb{N} : d \mid n\}$$

for its positive divisors, and we ask how complicated the first additive coincidences inside $\mathcal{D}(n)$ can be.

The smallest example already shows the phenomenon. We have

$$\mathcal{D}(6) = \{1, 2, 3, 6\}, \qquad 1 + 3 = 2 + 2.$$

So $\mathcal{D}(6)$ is not Sidon, although the divisor set of every proper divisor of 6 is. This motivates the minimality condition used throughout the paper: we want to study a collision at the point where it first appears under divisibility, rather than one inherited from a smaller divisor set.

The same collision can be measured in several natural ways. For an integer s, let

$$r_n(s) = \#\{\{x, y\} : x, y \in \mathcal{D}(n),\ x \le y,\ x + y = s\}, \qquad R(n) = \max_s r_n(s),$$

and define

$$\alpha(n) = \max\{|A| : A \subseteq \mathcal{D}(n),\ A \text{ is Sidon}\}$$

and

$$C(n) = \binom{|\mathcal{D}(n)| + 1}{2} - \big|\{x + y : x, y \in \mathcal{D}(n),\ x \le y\}\big| = \sum_s \max(r_n(s) - 1, 0).$$

These quantities give the four OEIS sequences studied together here. A398433 records the divisibility-minimal integers for which $R(n) > 1$; A398434 is $\alpha(n)$; A398435 is $C(n)$; and A398577 is $R(n)$. Thus the four sequences ask, respectively, *where* a collision first occurs, *how much* of the divisor set can be retained, *how many* repeated representations occur in total, and *how large* one collision can become. In particular,

$$R(n) = 1 \iff C(n) = 0 \iff \alpha(n) = |\mathcal{D}(n)|.$$

We call n *divisibility-minimal non-Sidon* if $R(n) > 1$ but every proper divisor $m \mid n$ satisfies $R(m) = 1$. The main proofs focus on A398577 at these minimal indices, because this is where higher multiplicity first appears without being inherited from a smaller divisor set. The families constructed later return to all four sequences simultaneously.

The central question is simple to state: *how many distinct prime factors are needed before a first collision can have multiplicity three, four, or more?* For squarefree minimal obstructions the first two thresholds are exact.

Theorem 1.1 (Threefold threshold). *Let n be squarefree and divisibility-minimal non-Sidon. If $R(n) \ge 3$, then*

$$\omega(n) \ge 5.$$

Equivalently, if $\omega(n) \le 4$, then $R(n) \le 2$.

Theorem 1.2 (Fourfold threshold). *Let n be squarefree and divisibility-minimal non-Sidon. If $R(n) \ge 4$, then*

$$\omega(n) \ge 6.$$

Equivalently, if $\omega(n) \le 5$, then $R(n) \le 3$.

These statements are not the result of a bounded search through small primes. The proof first turns a hypothetical repeated sum into a rigid factorization pattern. Each representation has a "defect", consisting of the prime factors of n that do not occur in either summand. Minimality forces the defects of different representations to be disjoint. What remains is a finite list of ways of distributing the prime factors between the two summands in each representation. Most of these possibilities are ruled out by elementary inequalities after the primes are written in terms of their nonnegative gaps; the remaining cases are disposed of by exact divisibility arguments or short algebraic contradictions. The computer is used to enumerate and certify this finite case analysis, not to sample numerical examples.

The two equivalent multiplicity bounds are sharp.

Corollary 1.3 (Exact sharpness levels)**.** *Among squarefree divisibility-minimal non-Sidon integers,*

$$\max_{\omega(n)\le 4} R(n) = 2, \qquad \max_{\omega(n)\le 5} R(n) = 3.$$

The first equality is witnessed by $n = 6$. For multiplicity three the first examples are much larger.

Theorem 1.4 (The two smallest threefold examples)**.** *The smallest and second-smallest squarefree divisibility-minimal non-Sidon integers with $R(n) = 3$ are*

$$n_1 = 81365669 = 7 \cdot 11 \cdot 31 \cdot 89 \cdot 383$$

and

$$n_2 = 82643495 = 5 \cdot 11 \cdot 41 \cdot 67 \cdot 547.$$

Their unique sums of multiplicity three are, respectively,

$$2770 = 11 + 31 \cdot 89 = 89 + 7 \cdot 383 = 383 + 7 \cdot 11 \cdot 31$$

and

$$2802 = 5 \cdot 11 + 41 \cdot 67 = 5 \cdot 11 \cdot 41 + 547 = 67 + 5 \cdot 547.$$

At first sight these two identities look unrelated. In fact they come from the same three-step linear cycle. We solve

$$x_1 + r_2x_2 = x_2 + r_3x_3 = x_3 + r_1x_1$$

in positive integers and then specialize the solution in two different ways. This produces the divisor identities

$$b + cd = d + ae = e + abc$$

and

$$ab + cd = abc + e = d + ae.$$

The two templates lead to four explicit polynomial families. Whenever the four variable factors in one of these families are prime (apart from one stated exceptional parameter), the resulting five-prime integer is a divisibility-minimal example with $R(n) = 3$. Under Schinzel's Hypothesis H, each family contains infinitely many such integers.

The paper therefore has two complementary parts. The first, sections 2 to 6, establishes the sharp multiplicity bounds for at most four and five prime factors. The second, sections 7 to 11, asks why the sharp five-prime examples have the identities displayed above and turns that explanation into parametric families. Section 12 then translates the resulting family theorems back into simultaneous statements about all four OEIS sequences. The computational ingredients are described where they enter the proof, while section 13 collects the reproducibility details. The full elementary closure of the 36 residual threefold orientations is recorded in Appendix A.

Recent work of Letendre concerns additive relations of the form $d_1 + d_2 = d_3$ among divisors and gives quantitative bounds for their number [2]. Our problem is different: the equalities here have the form $d_1 + d_2 = d_3 + d_4$, and the emphasis is on representation multiplicity under divisibility-minimality.

2 Definitions and elementary facts

We use unordered pairs with repetition allowed. In particular, a relation such as $1+3 = 2+2$ is a genuine Sidon collision.

Definition 2.1. For $n \geq 1$ and $s \in \mathbb{Z}$, set

$$r_n(s) = \#\{\{x, y\} : x, y \in \mathcal{D}(n),\ x \leq y,\ x + y = s\}, \qquad R(n) = \max_s r_n(s).$$

A positive integer n is *divisibility-minimal non-Sidon* if $R(n) > 1$ and $R(m) = 1$ for every proper divisor $m \mid n$.

Example 2.2. For $n = 6$ the repeated sum

$$4 = 1 + 3 = 2 + 2$$

shows that $R(6) = 2$. The proper divisors $1, 2, 3$ have Sidon divisor sets, so 6 is divisibility-minimal non-Sidon. By contrast, 12 is not minimal: its divisor set has several collisions, but it already contains $\mathcal{D}(6)$.

A useful simplification is that minimality need only be checked one prime at a time.

Lemma 2.3. *Let n be squarefree. Then n is divisibility-minimal non-Sidon if and only if $R(n) > 1$ and*

$$R(n/p) = 1$$

for every prime $p \mid n$.

Proof. The forward implication is immediate. Conversely, every proper divisor m of a squarefree n omits some prime $p \mid n$. Hence $m \mid n/p$ and $\mathcal{D}(m) \subseteq \mathcal{D}(n/p)$. Every subset of a Sidon set is Sidon. $\square$

Throughout the two threshold proofs, n is squarefree and $\omega(n)$ denotes its number of distinct prime factors.

3 The core–defect decomposition

The main structural observation is easiest to describe informally. Suppose the same integer M is represented several times as a sum of two divisors of a squarefree minimal obstruction n. Look at the prime factors of n that are absent from a given pair of summands. Those missing factors form the *defect* of that representation. Two different representations cannot miss the same prime: if they did, both collisions would already live inside the divisor set of a proper divisor of n. Thus the defects are pairwise coprime, while the prime factors that occur outside all defects form a common *core*.

We now make this precise. Suppose

$$M = x_i + y_i, \qquad i = 1, \dots, r,$$

are distinct representations, with $x_i, y_i \mid n$ and $x_i \le y_i$. Put

$$m_i = \operatorname{lcm}(x_i, y_i), \qquad h_i = \frac{n}{m_i}.$$

Thus h_i is exactly the part of n missing from the ith representation.

Lemma 3.1 (Full support)**.** *If n is divisibility-minimal non-Sidon, then for distinct representations $i \ne j$,*

$$\operatorname{lcm}(m_i, m_j) = n.$$

Proof. All four summands in representations i and j divide $\operatorname{lcm}(m_i, m_j)$. If this least common multiple were a proper divisor of n, those two representations would already give a Sidon collision in a smaller divisor set, contrary to minimality. ☐

Proposition 3.2 (Core–defect decomposition)**.** *Let n be squarefree and divisibility-minimal non-Sidon. Suppose that one sum has $r \ge 3$ distinct representations. Then there are pairwise coprime squarefree integers*

$$\kappa, h_1, \dots, h_r$$

such that

$$n = \kappa h_1 \cdots h_r$$

and, for every i,

$$\gcd(x_i, y_i) = 1, \qquad x_i y_i = \frac{n}{h_i} = \kappa \prod_{j \ne i} h_j.$$

Moreover, at most one of the defects h_i is equal to 1*.*

Proof. By lemma 3.1, the defects are pairwise coprime. Indeed, if a prime divided both h_i and h_j, it would be absent from both m_i and m_j, contradicting $\operatorname{lcm}(m_i, m_j) = n$.

It remains to see that the two summands in each representation are coprime. Suppose a prime p divided both x_i and y_i. Then $p \mid M$. Fix another representation $M = x_j + y_j$. The prime p cannot divide exactly one of x_j, y_j, since their sum is divisible by p. If it divided both, then

$$M/p = x_i/p + y_i/p = x_j/p + y_j/p$$

would be a collision inside $\mathcal{D}(n/p)$: all four quotients divide n/p. The two unordered quotient pairs remain distinct, because equality after division would imply $\{x_i, y_i\} = \{x_j, y_j\}$ after multiplication by p. Since n/p is a proper divisor of n, this contradicts divisibility-minimality. Consequently p divides neither x_j nor y_j for every $j \ne i$. Therefore p belongs to every defect h_j with $j \ne i$. Since $r \ge 3$, it would lie in at least two defects, contrary to their pairwise coprimality.

Thus $\gcd(x_i, y_i) = 1$. Since n is squarefree, this gives

$$m_i = x_i y_i = \frac{n}{h_i}.$$

The pairwise coprime defects have a product dividing n; define

$$\kappa = \frac{n}{h_1 \cdots h_r}.$$

This yields the stated factorization.

Finally, two defects cannot both be 1. If $h_i = h_j = 1$, then the corresponding unordered pairs have the same sum M and the same product n. They are therefore the two roots of the same quadratic $T^2 - MT + n$ and must coincide, contrary to the assumption that the representations are distinct. □

We call κ the *core*. A convenient summary of a configuration is its *core–defect type*

$$(\omega(\kappa); \omega(h_1), \ldots, \omega(h_r)).$$

Only the number of prime factors in each part is recorded. When $\omega(n)$ is bounded, there are only finitely many such types.

There is one further simplification in the even case.

Lemma 3.3 (Parity of the core)**.** *If n is even in the setting of proposition 3.2, then $2 \mid \kappa$. In particular, no defect is even.*

Proof. Suppose $2 \mid h_i$. Then $x_i y_i = n/h_i$ is odd, so $x_i + y_i$ is even. The defects are pairwise coprime, so no other defect contains 2. Hence for every $j \neq i$, the product $x_j y_j = n/h_j$ is even. Since x_j and y_j are coprime, exactly one is even, making $x_j + y_j$ odd. The same sum M cannot be both even and odd. Thus 2 lies in the core.

□

Lemma 3.4 (Complete core–defect type lists)**.** *Let $m = \omega(n)$ and sort the defect sizes in nonincreasing order. In the threefold problem with $m \leq 4$, the possible types are exactly*

case	*types*
odd $m = 2$	$(0; 1, 1, 0)$
odd $m = 3$	$(0; 1, 1, 1)$, $(1; 1, 1, 0)$, $(0; 2, 1, 0)$
even $m = 3$	$(1; 1, 1, 0)$
odd $m = 4$	$(1; 1, 1, 1)$, $(0; 2, 1, 1)$, $(2; 1, 1, 0)$, $(1; 2, 1, 0)$, $(0; 3, 1, 0)$, $(0; 2, 2, 0)$
even $m = 4$	$(1; 1, 1, 1)$, $(2; 1, 1, 0)$, $(1; 2, 1, 0)$.

In the fourfold problem with $m \leq 5$, the possible types are exactly

case	*types*
odd $m = 3$	$(0; 1, 1, 1, 0)$
odd $m = 4$	$(0; 1, 1, 1, 1)$, $(0; 2, 1, 1, 0)$, $(1; 1, 1, 1, 0)$
even $m = 4$	$(1; 1, 1, 1, 0)$
odd $m = 5$	$(1; 1, 1, 1, 1)$, $(0; 2, 1, 1, 1)$, $(0; 3, 1, 1, 0)$
	$(0; 2, 2, 1, 0)$, $(1; 2, 1, 1, 0)$, $(2; 1, 1, 1, 0)$
even $m = 5$	$(1; 1, 1, 1, 1)$, $(1; 2, 1, 1, 0)$, $(2; 1, 1, 1, 0)$.

Proof. Write a type as $(k; h_1, \ldots, h_r)$, where $k = \omega(\kappa)$ and $h_i = \omega(h_i)$; then

$$k + h_1 + \cdots + h_r = m.$$

By proposition 3.2, at most one defect is empty, so at least $r - 1$ of the h_i are positive. By lemma 3.3, in the even case $k \geq 1$. The displayed lists are therefore just the integer partitions of $m - k$ into r parts with at most one zero, sorted in nonincreasing order. Enumerating those partitions for $r = 3$, $m \leq 4$, and for $r = 4$, $m \leq 5$, gives exactly the lists above. □

4 From the structural reduction to a finite problem

The finite part of the proof has three layers. First, the core–defect decomposition gives a complete finite list of combinatorial types and orientations. Second, exact sign certificates eliminate the generic orientations throughout the ordered-prime region. Third, the remaining finite residual sets are closed by modular certificates or by explicit elementary arguments. The purpose of this section is to make each of these layers precise before the threshold theorems are assembled in sections 5 and 6.

The preceding proposition replaces an arbitrary collection of divisor pairs by a finite combinatorial problem. Fix a core–defect type and the actual primes assigned to it. In representation i the product of the two summands is

$$P_i = \kappa \prod_{j \neq i} h_j.$$

Because P_i is squarefree and the summands are coprime, choosing the unordered pair $\{u_i, v_i\}$ with $u_i v_i = P_i$ amounts simply to deciding, for each prime dividing P_i, which of the two factors receives it. If P_i contains q_i primes, there are therefore exactly $2^{q_i - 1}$ unordered coprime factorizations.

We call a simultaneous choice of one factorization for every representation an *orientation.*

Lemma 4.1 (Orientation bijection)**.** *Fix an ordered prime tuple and a core–defect type from lemma 3.4. For labeled representations, the candidate collision configurations compatible with proposition 3.2 are in bijection with the block assignments and unordered coprime factorizations generated in this section. In the fourfold audit, after quotienting only by permutation of the four representation labels, the canonical convention of lemma 4.3 gives exactly one generated orientation for each orbit.*

Proof. Given a candidate collision, the core and defects are determined by proposition 3.2. Each prime of the ordered tuple therefore belongs to exactly one of the disjoint blocks $\kappa, h_1, \ldots, h_r$. For representation i, the product of its two summands is $P_i = n/h_i$, and squarefreeness together with $\gcd(x_i, y_i) = 1$ says that the unordered pair $\{x_i, y_i\}$ is exactly a bipartition of the prime factors of P_i. This recovers one generated block assignment and one generated unordered factorization for every representation.

Conversely, a generated block assignment and a choice of one unordered coprime factorization of every P_i produce divisors of n with precisely the core–defect support prescribed by proposition 3.2. Thus the two constructions are inverse before the equal-sum equations are imposed. In the fourfold case, permuting representation labels changes only the ordering of the four defect blocks and their four factor pairs; it does not change any ordered prime variable. The canonical sorting rule therefore selects one representative of each such orbit. □

Once an orientation is fixed, the condition that all sums agree is an explicit polynomial system. For three representations write

$$S_i = u_i + v_i, \qquad F_1 = S_2 - S_1, \qquad F_2 = S_3 - S_1;$$

for four representations add $F_3 = S_4 - S_1$. A collision is possible only if all the relevant F_i vanish.

Lemma 4.2 (Orientation count with labeled representations)**.** *Consider a type $(k; h_1, \ldots, h_r)$ with $m = k + h_1 + \cdots + h_r$ prime factors, and temporarily regard the r representations as* *labeled. If n is odd, the number of assignments of the ordered primes to the core and the labeled defects is*

$$A_{\text{odd}} = \frac{m!}{k!\,h_1!\cdots h_r!}.$$

If n is even, then 2 is fixed in the core and the number is

$$A_{\text{even}} = \frac{(m-1)!}{(k-1)!\,h_1!\cdots h_r!}.$$

For every such block assignment the number of orientations is

$$\prod_{i=1}^{r} 2^{m-h_i-1}.$$

Hence the total labeled orientation count is the corresponding product of these two factors.

Proof. The multinomial coefficients count the assignments of the m distinct ordered primes to the core and the labeled defect blocks. In the even case the prime 2 has already been placed in the core, leaving $m-1$ odd primes and a remaining core size $k-1$. Representation i contains every prime except those in h_i, hence exactly $m-h_i$ primes. Fixing one of these primes on one side of the factor pair removes the complement symmetry and leaves exactly 2^{m-h_i-1} unordered factorizations. The choices for different representations are independent. □

The threefold audit keeps the representations labeled, so lemma 4.2 reproduces every number in table 1. For example, for the odd type $(1;1,1,1)$ there are

$$\frac{4!}{1!^4} = 24$$

block assignments, and each of the three products contains three primes and therefore has $2^{3-1} = 4$ unordered factorizations. The count is

$$24 \cdot 4^3 = 1536.$$

For $(0;2,1,1)$ there are $4!/(2!1!1!) = 12$ assignments, while the three products contain $2,3,3$ primes; hence

$$12 \cdot 2 \cdot 4 \cdot 4 = 384.$$

The remaining rows are obtained by the same formula, and their displayed sum is 6092.

For the fourfold audit we remove the redundant relabeling of the four representations already at the block-assignment stage.

Lemma 4.3 (Canonical fourfold block convention)**.** *Fix the ordered prime list. In a fourfold type, order the defect blocks first by decreasing cardinality and, among blocks of the same size, lexicographically by their lists of prime indices; put the unique empty defect last. This chooses exactly one representative of every permutation orbit of the four representations. No prime variable is permuted.*

Proof. Let X be the set of fully labeled fourfold block assignments whose four defect sizes form the prescribed multiset. The group S_4 acts on X by relabeling the representations, hence by permuting the four defects and the four equations $S_i = M$. This action is free. Indeed, if a permutation fixes a labeled block assignment, then it sends every defect block to itself. Distinct nonempty defects are disjoint and therefore cannot be equal, while proposition 3.2

allows at most one empty defect. Thus every defect block is distinct as an actual subset of the ordered prime list, and the only stabilizer element is the identity.

Now first put the defect sizes in decreasing order. Within one S_4-orbit this removes all permutations between blocks of different sizes. If a size s occurs with multiplicity m_s, exactly $m_s!$ assignments in the resulting size-ordered slice remain, corresponding to the permutations of the m_s distinct blocks having that common size. Lexicographic ordering of their prime-index lists selects exactly one of these $m_s!$ possibilities. Doing this independently for every repeated size therefore selects exactly one point of each original S_4-orbit. Equivalently, starting from the multinomial count with size-labeled slots, one divides by precisely $\prod_s m_s!$ and by no additional stabilizer factor.

At no stage is an ordered prime variable moved: the action changes only representation labels and the associated defect slots. Hence the canonicalization preserves the ordered-prime cone. □

If m_s denotes the multiplicity with which the defect size s occurs, the number of canonical fourfold block decompositions is therefore

$$B_{\text{odd}} = \frac{m!}{k! \prod_i h_i! \prod_s m_s!}, \qquad B_{\text{even}} = \frac{(m-1)!}{(k-1)! \prod_i h_i! \prod_s m_s!}.$$

The extra factors $m_s!$ are exactly the permutations of equal-sized defect blocks removed by the canonical convention.

The block-decomposition counts used later are consequently elementary. For example, in odd type $(1; 1, 1, 1, 1)$ one chooses the core prime in 5 ways and the four singleton defects are then forced by the canonical convention. Each representation contains four primes, giving $8^4 = 4096$ factorization choices, hence $5 \cdot 4096 = 20480$ orientations. In type $(0; 2, 2, 1, 0)$ one chooses the singleton defect in 5 ways and partitions the remaining four primes into two unordered pairs in 3 ways, giving 15 canonical block decompositions; each decomposition has 2048 factorization choices, hence 30720 orientations.

The computer-assisted part of the proof is an exhaustive enumeration of these precisely defined orientations. Each rejection comes with an exact certificate. Two types of certificate do almost all of the work.

4.1 Sign certificates

Let $p_1 < \cdots < p_q$ be odd primes. Write their gaps as

$$p_1 = 3 + 2z_1, \qquad p_i = p_{i-1} + 2 + 2z_i \quad (i \geq 2), \qquad z_i \geq 0.$$

In an even case we set the first prime equal to 2 and parametrize the remaining odd primes in the same way. A small primitive integral linear combination

$$G = \mu_1 F_1 + \cdots + \mu_k F_k$$

is expanded in the independent variables z_i. If every coefficient is nonnegative and the constant term is positive, then $G > 0$ throughout the whole ordered-prime region, so the orientation cannot be a solution. The same argument with $-G$ covers the negative case.

The implementation uses an injective Kronecker encoding of the multivariate polynomial for speed in the five-prime audit, but the mathematical certificate is only the fixed sign of an explicitly expanded polynomial. In the threefold audit every accepted certificate is also rechecked directly in the original multivariate gap variables.

Example. Take the odd threefold type $(0;1,1,1)$, with core 1 and defects $a < b < c$. One orientation gives

$$S_1 = b + c, \qquad S_2 = a + c, \qquad S_3 = ab + 1.$$

Then $F_1 = a - b$. With

$$a = 3 + 2z_1, \qquad b = 5 + 2z_1 + 2z_2,$$

we obtain

$$-F_1 = b - a = 2 + 2z_2 > 0.$$

So this orientation is impossible for every ordered choice of the primes, not merely for primes below some numerical bound.

4.2 Modular certificates

A small number of orientations survive the sign test. Their second-stage certificates have a fixed, machine-readable form. Let $D = (F_1, F_2, F_3)$ be the three difference polynomials of a fourfold orientation. A stored modular certificate consists of

$$(c_1, c_2, c_3), \qquad p, \qquad g = (c_1F_1 + c_2F_2 + c_3F_3)|_{p=0}\,,$$

where (c_1, c_2, c_3) is a primitive integer triple from a fixed finite search box, p is one of the ordered prime variables, and g is supplied with its exact factorization over $\mathbb{Z}$.

Lemma 4.4 (Validity of a modular certificate)**.** *Suppose the factorization is*

$$g = u \prod_j f_j^{e_j}.$$

Assume $|u| < p$ and, for every factor f_j, either f_j is a monomial in prime variables different from p, or the ordered-prime inequalities certify

$$0 < |f_j| < p.$$

Then the orientation has no solution in ordered distinct primes.

Proof. If all F_i vanished at the prime tuple, then the integer $c_1F_1 + c_2F_2 + c_3F_3$ would vanish. Reducing this equality modulo the numerical prime p gives

$$g \equiv 0 \pmod{p},$$

because substituting $p = 0$ is exactly reduction of the polynomial modulo p. The unit u is not divisible by p because $|u| < p$. A monomial in the other distinct prime variables is also not divisible by p, and a nonzero integer of absolute value $< p$ is not divisible by p. Hence no factor on the right is divisible by p, contradicting $p \mid g$. □

For every modular closure we store the case index, the coefficient triple, the modulus prime variable, and the exact factorization of g. These records make each rejection directly reconstructible. The complete ledgers, independent record checkers, and reproduction commands are described together in section 13; here we keep the focus on the mathematical form of the certificates.

Theorem 4.5 (Finite certificate theorem)**.** *The following finite statements are established by exhaustive certified enumeration over the domains specified in lemmas 3.4 and 4.1 to 4.3.*

(i) *In the threefold problem with at most four prime factors, the complete orientation set has size* 6092*;* 6056 *orientations have sign certificates and the remaining* 36 *form the* 16 *systems closed in Appendix A.*

(ii) *In the even five-prime fourfold problem, the complete orientation set has size* 61440*; the sign stage leaves* 38 *orientations. Exactly* 37 *have modular certificates satisfying lemma 4.4, and the remaining residual (case* 31 *in the independent v8 ledger) is impossible by the elementary argument in section 6.1.*

(iii) *In the odd five-prime fourfold problem, the complete orientation set has size* 296960*; the sign stage leaves exactly* 509 *orientations. Exactly* 498 *of these have stored modular certificates satisfying lemma 4.4; the remaining indices are*

$$33, 147, 148, 169, 239, 330, 347, 423, 426, 455, 456,$$

and all eleven are closed by the explicit arguments in section 6.

Proof. The domains being enumerated are fixed independently of the programs by lemmas 3.4 and 4.1 to 4.3. The exact audits enumerate those finite domains and assert the row counts and residual counts stated above.

For the odd five-prime problem, the complete list of 509 sign-stage residual systems is stored explicitly; 498 carry modular certificates satisfying lemma 4.4, and a separate record checker verifies those certificates without searching for new ones. The complementary eleven indices are exactly the cases closed in section 6. For the even five-prime problem, an independent reconstruction produces the same 38 residual systems, verifies modular certificates for 37, and leaves the single elementary case treated in section 6.1. For the threefold problem, an independent implementation reconstructs the 6092 orientations, the 6056 strict sign closures, and the 36 residual orientations; those 36 collapse to the 16 equation groups proved impossible in Appendix A.

Thus every finite remainder is accounted for either by an exact stored certificate or by an explicit argument in the manuscript. The corresponding files and verification commands are collected in section 13. No bounded prime search enters either threshold theorem. □

A complete stored certificate. Certificate record 1 of the ancillary JSON ledger belongs to the odd five-prime type $(1; 1, 1, 1, 1)$, with core prime a and defects $b < c < d < e$. Its four sums are

$$S_1 = ad + ce, \quad S_2 = ab + de,$$
$$S_3 = abe + c, \quad S_4 = abcd + 1.$$

The stored record is

$$(c_1, c_2, c_3) = (1, 0, 0), \qquad p = e, \qquad g = a(b - d) = 1 \cdot a \cdot (b - d).$$

Indeed, taking $G = S_2 - S_1$ and reducing modulo e gives exactly this g. The factor a is a monomial in a prime different from e, while $b - d < 0$ and

$$0 < d - b < e.$$

Thus both factors satisfy lemma 4.4. If all four sums were equal, then $e \mid a(b - d)$, impossible. The verifier performs these same exact checks on all 498 stored records.

The few systems that survive both uniform tests are handled individually by elementary algebra, parity, residue classes, or the known order of the primes. All eleven nonuniform odd five-prime closures are printed in section 6.

4.3 Grouping genuinely equivalent threefold residuals

For the threefold audit, the remaining orientations can be grouped without changing the ordered-prime domain. Within each representation we may interchange the two summands, and we may permute the three representations themselves. Sorting these data gives a canonical equation key.

Lemma 4.6 (Safe residual grouping)**.** *Two threefold residual orientations with the same canonical equation key impose exactly the same unordered system of three sum equations. Passing to the key does not permute any prime variable and therefore preserves the ordered region* $p_1 < \cdots < p_m$*.*

Proof. Interchanging u_i and v_i does not change the sum $S_i = u_i + v_i$, and permuting the three representation labels only permutes the equations $S_1 = M, S_2 = M, S_3 = M$. The canonical key first sorts the two monomials inside each sum and then sorts the three sums. No substitution $p_i \leftrightarrow p_j$ is performed. Hence all orientations with the same key are literally the same polynomial equality system on the same ordered prime variables. □

For example, in the even type $(1; 1, 1, 1)$ with core prime $a = 2$ and defects b, c, d, the factor pairs

$$(ac, d), \qquad (ad, b), \qquad (a, bc)$$

lead to

$$a + bc = ac + d = ad + b.$$

Six orientations give this same equation system. Altogether the threefold sign audit leaves 36 orientations, which collapse to the 16 equation groups listed and proved impossible in Appendix A. In particular, this grouping uses no permutation symmetry of the ordered prime variables.

5 The exact threefold threshold

We can now prove theorem 1.1. Suppose that a common sum has three distinct representations and that n has at most four prime factors. By proposition 3.2, the primes of n are distributed between one core and three pairwise coprime defects, with at most one empty defect. Up to ordering the defect sizes, this leaves only the types in table 1.

The type list is exhaustive by lemma 3.4, and every orientation count in the fourth column follows from lemma 4.2. Thus the total 6092 is a combinatorial identity. The exact sign audit closes 6056 orientations and leaves 36; under the safe grouping of lemma 4.6, these become 16 equation systems. Appendix A gives an elementary proof for every one of them. An independent implementation reproduces the same audit; the verification details are collected in section 13.

There is one ordered prime solution among the residual equations:

$$(a, b, c, d) = (2, 3, 5, 7).$$

It corresponds to $n = 210$, but it is not a minimal obstruction because $6 \mid 210$ and $\mathcal{D}(6)$ already has the collision $1 + 3 = 2 + 2$.

Table 1: Exact threefold orientation audit. A type is $(\omega(\kappa); \omega(h_1), \omega(h_2), \omega(h_3))$.

Parity	$\omega(n)$	Type	Orientations	Sign-closed	Residual
odd	2	$(0;1,1,0)$	4	4	0
odd	3	$(0;1,1,1)$	48	48	0
odd	3	$(1;1,1,0)$	96	96	0
odd	3	$(0;2,1,0)$	24	24	0
even	3	$(1;1,1,0)$	32	32	0
odd	4	$(1;1,1,1)$	1536	1536	0
odd	4	$(0;2,1,1)$	384	378	6
odd	4	$(2;1,1,0)$	1536	1520	16
odd	4	$(1;2,1,0)$	768	764	4
odd	4	$(0;3,1,0)$	128	128	0
odd	4	$(0;2,2,0)$	192	192	0
even	4	$(1;1,1,1)$	384	378	6
even	4	$(2;1,1,0)$	768	764	4
even	4	$(1;2,1,0)$	192	192	0
		Total	6092	6056	36

Proof of theorem 1.1. If $R(n) \geq 3$ and $\omega(n) \leq 4$, choose three representations of a common sum and apply the preceding exhaustive classification. Every orientation is impossible for a divisibility-minimal n: the only prime solution is the nonminimal case $n = 210$. Hence a squarefree divisibility-minimal non-Sidon integer with $R(n) \geq 3$ must have at least five prime factors. □

6 The exact fourfold threshold

For four representations there are four defects, and at least three of them are nontrivial. The complete type list is the fourfold part of lemma 3.4. We use the canonical block convention of lemma 4.3; hence the block-decomposition and orientation counts below can be checked directly before any symbolic certificate is considered.

6.1 Small and even cases

The configurations with at most four prime factors are disposed of entirely by sign certificates. For an even five-prime integer, 38 orientations survive that first stage. Of these, 37 admit modular certificates and one short residual remains to be closed by hand. In the table, B is the number of canonical core–defect block decompositions and F is the product of the unordered factorization counts for one decomposition, so that the orientation count is BF.

For example, in the even type $(1;2,1,1,0)$ the prime 2 is the core, the two-prime defect can be chosen from the four odd primes in $\binom{4}{2} = 6$ ways, and the remaining singleton defects are then canonically ordered. The four representation products contain $3, 4, 4, 5$ primes, so

$$F = 2^2\, 2^3\, 2^3\, 2^4 = 4096,$$

and the orientation count is $6 \cdot 4096 = 24576$. The other rows are checked identically.

By theorem 4.5(ii), 37 of the 38 residual even five-prime orientations have modular certificates. The remaining system is short enough to close directly.

Table 2: Small and even fourfold audit, with the combinatorial orientation counts exposed.

Case	B	F	Orientations	Residual after sign test
Odd 3: $(0;1,1,1,0)$	1	32	32	0
Odd 4: $(0;1,1,1,1)$	1	256	256	0
Odd 4: $(0;2,1,1,0)$	6	256	1536	0
Odd 4: $(1;1,1,1,0)$	4	512	2048	0
Even 4: $(1;1,1,1,0)$	1	512	512	0
		Small total	4384	0
Even 5: $(1;1,1,1,1)$	1	4096	4096	2
Even 5: $(1;2,1,1,0)$	6	4096	24576	14
Even 5: $(2;1,1,1,0)$	4	8192	32768	22
		Even five-prime total	61440	38

The single nonuniform even residual. For case 31 of the v8 residual ledger the four oriented sums are

$$S_1 = 2c + de, \qquad S_2 = 2be + d, \qquad S_3 = 2bcd + 1, \qquad S_4 = 2bd + ce,$$

where $2 < b < c < d < e$ are primes. From $S_1 = S_3$ we obtain

$$d(2bc - e) = 2c - 1.$$

The right-hand side is positive and satisfies $0 < 2c - 1 < 2d$. Since it is divisible by the odd prime d, necessarily

$$d = 2c - 1, \qquad 2bc - e = 1.$$

On the other hand, $S_1 = S_2$ gives

$$e(d - 2b) = d - 2c.$$

Substituting $d = 2c - 1$ yields $e(d - 2b) = -1$, impossible because $e > 1$ and $d - 2b$ is an integer. Thus the final even residual is impossible.

6.2 Odd five-prime case

Now write the five primes as $a < b < c < d < e$. The six possible types are exhaustive by lemma 3.4. Their canonical block counts and exact first-stage counts are as follows.

Table 3: Odd five-prime fourfold audit, with explicit block and factorization counts.

Type	B	F	Orientations	Residual after sign test
$(1;1,1,1,1)$	5	4096	20480	53
$(0;2,1,1,1)$	10	2048	20480	127
$(0;3,1,1,0)$	10	2048	20480	15
$(0;2,2,1,0)$	15	2048	30720	22
$(1;2,1,1,0)$	30	4096	122880	187
$(2;1,1,1,0)$	10	8192	81920	105
		Total	296960	509

As examples, the first row has five possible core primes and $8^4 = 4096$ factorization choices. For $(0;2,2,1,0)$ there are 5 choices for the singleton defect and 3 unordered partitions of the remaining four primes into two pairs, giving $B = 15$; the product of the four factorization counts is $F = 2048$. Thus every entry in the orientation column, and hence the total 296960, follows directly from the combinatorics.

The sign test leaves 509 orientations. By theorem 4.5(iii), 498 have modular certificates. The remaining indices are exactly

$$33, 147, 148, 169, 239, 330, 347, 423, 426, 455, 456.$$

We now close these eleven cases explicitly. The verification package also checks the displayed algebraic reductions; the decisive inequalities and divisibility arguments are given here.

Case 33

The four sums include

$$bc + de, \qquad acd + e, \qquad abe + d, \qquad bcd + a.$$

Equality of the second and fourth gives

$$e - a = cd(b - a),$$

so $e > cd$. Equality of the second and third gives

$$e(ab - 1) = d(ac - 1).$$

Since $ac - 1 < c(ab - 1)$, this implies $e < cd$, a contradiction.

Case 147

The relevant equalities are

$$bcd + 1 = acd + e = ab + de.$$

They give

$$e - 1 = cd(b - a), \qquad d(bc - e) = ab - 1.$$

Thus $e > cd$ and $e < bc < cd$, a contradiction.

Case 148

Here

$$bcd + 1 = acd + e = ab + ce,$$

so

$$e - 1 = cd(b - a), \qquad c(bd - e) = ab - 1.$$

Hence $e > cd$ and $e < bd < cd$, again a contradiction.

Case 169

The relevant equalities are

$$bd + ce = ac + de = acd + b.$$

The first gives

$$d(e - b) = c(e - a).$$

Since $(c, d) = 1$, there is a positive integer L such that

$$e - b = cL, \qquad e - a = dL.$$

Since e and b are odd, $e - b$ is even. As c is odd and $e - b = cL$, the integer L is even, hence $L \geq 2$. The second displayed equality of sums gives

$$d(ac - e) = ac - b.$$

Reducing this congruence modulo c and using $e \equiv b \pmod{c}$ yields $d \equiv 1 \pmod{c}$. Since c and d are odd primes with $d > c$, writing $d = 1 + kc$ forces k to be even and therefore

$$d - c \geq c + 1.$$

Subtracting $e - b = cL$ from $e - a = dL$ now gives

$$b - a = (d - c)L \geq 2(c + 1) > c,$$

contrary to $b - a < c$.

Case 239

The equalities

$$ac + de = acd + b, \qquad abe + 1 = ac + de$$

give

$$d(ac - e) = ac - b, \qquad e(ab - d) = ac - 1.$$

Put $m = ac - e$. Then

$$0 < m = \frac{ac - b}{d} < a < e.$$

Since $ac - 1 = e + m - 1 < 2e$, the second identity gives

$$0 < ab - d < 2.$$

Thus $ab - d = 1$, and then $e = ac - 1$, so $ac - e = 1$. This is impossible because $ac - e$ is the difference of two odd integers and hence even.

Case 330

The reductions are

$$e(ab - c) = d(ab - 1), \qquad d(bc - e) = ac - 1.$$

The first gives $e \mid ab - 1$. The second gives

$$bc - e = \frac{ac - 1}{d} < a,$$

so

$$e > bc - a > ab - 1,$$

where the last inequality follows from

$$bc - a - (ab - 1) = b(c - a) - a + 1 > 0.$$

This contradicts $e \mid ab - 1$.

Cases 347, 455, and 456

In all three cases the relevant equalities reduce to

$$e(ab-c) = d(ab-1), \qquad d(e-1) = b(ae-c).$$

Hence

$$e \mid ab-1, \qquad b \mid e-1.$$

Write $ab-1 = eq$. Since $e > b$, we have $0 < q < a < b$. Modulo b, however, $e \equiv 1$, and hence $q \equiv -1 \pmod b$, impossible for $0 < q < a < b$.

Case 423

Here the equalities imply

$$e(ab-c) = d(ab-1), \qquad ab(cd-e) = d-1.$$

Thus $e \mid ab-1$ and $ab \mid d-1$. But $d < e \le ab-1$, so

$$0 < d-1 < ab,$$

contradicting $ab \mid d-1$.

Case 426

Here

$$e \mid ab-1, \qquad a \mid e-1.$$

Write $ab-1 = eq$. Since $e > b$, $0 < q < a$. Reducing modulo a gives $q \equiv -1 \pmod a$, hence $q = a-1$. Therefore

$$e - b = \frac{b-1}{a-1} < b.$$

A further equality gives

$$d(ac-e) = ac-b.$$

Let $m = ac-e$. Then m is a positive even integer, so $m \ge 2$, and subtraction yields

$$e - b = (d-1)m \ge 2(d-1) > b,$$

a contradiction.

This proves that all 509 sign-stage residual orientations are impossible: 498 by uniform modular certificates and the final eleven by the explicit arguments above.

Proof of theorem 1.2. Suppose $R(n) \ge 4$ and $\omega(n) \le 5$. The complete type list is given by lemma 3.4. The small cases in table 2 exclude $\omega(n) \le 4$. Thus $\omega(n) = 5$. If n is even, theorem 4.5(ii) eliminates the 38 sign-stage residuals (37 modular certificates and the single elementary closure above). If n is odd, theorem 4.5(iii) together with the eleven explicit closures above eliminates every canonical orientation. Hence $\omega(n) \ge 6$. □

Proof of corollary 1.3. Theorems 1.1 and 1.2 give the upper bounds. The integer 6 is divisibility-minimal non-Sidon and has $R(6) = 2$ because $1+3 = 2+2$. The examples in theorem 1.4 have $R(n) = 3$. □

7 A cyclic collision lemma

The threshold theorem tells us that multiplicity three first becomes possible with five prime factors, but it does not explain why the two smallest examples have the particular identities displayed in theorem 1.4. Both identities are instances of the same elementary linear cycle.

Imagine three quantities x_1, x_2, x_3 arranged cyclically, with fixed positive multipliers r_1, r_2, r_3. Requiring the three expressions

$$x_1 + r_2x_2, \qquad x_2 + r_3x_3, \qquad x_3 + r_1x_1$$

to be equal gives two independent homogeneous linear equations in three unknowns. Hence one expects a one-dimensional family, and the following lemma writes its primitive integer generator explicitly.

Lemma 7.1 (Cyclic collision lemma)**.** *Let r_1, r_2, r_3 be positive integers and put*

$$A = r_2(r_3 - 1) + 1, \qquad B = r_3(r_1 - 1) + 1, \qquad C = r_1(r_2 - 1) + 1,$$

and $g = \gcd(A, B, C)$. The positive integer solutions of

$$x_1 + r_2x_2 = x_2 + r_3x_3 = x_3 + r_1x_1$$

are exactly

$$(x_1, x_2, x_3) = m\left(\frac{A}{g}, \frac{B}{g}, \frac{C}{g}\right), \qquad m \in \mathbb{N}.$$

The common sum is

$$\frac{m(1 + r_1r_2r_3)}{g}.$$

Proof. Subtracting consecutive equalities gives two homogeneous linear equations. Their rational solution space is one-dimensional. A direct calculation shows that (A, B, C) lies in this nullspace because

$$A + r_2B = B + r_3C = C + r_1A = 1 + r_1r_2r_3.$$

Dividing by g gives the primitive positive integer vector, and every positive integer solution is a positive multiple of it. □

The usefulness of the lemma is that the coefficients r_i can themselves be chosen as products of primes. The three coordinates then become candidate prime factors, and divisibility conditions on g turn into the auxiliary factorizations that generate the two templates below.

8 The two smallest sharp examples

Before developing the parametric families, we record how the two smallest examples were certified. This is the one place where a bounded exhaustive search is used. It proves a statement about the *smallest* examples; the threshold theorems themselves are independent of any numerical search bound.

For every squarefree $n \leq 82643495$ with $5 \leq \omega(n) \leq 8$, a C++17 program enumerates strictly increasing prime tuples by recursion. At each recursion node it multiplies by the smallest possible remaining primes; if that lower bound already exceeds 82643495, the branch is stopped. Thus every squarefree product in the stated range occurs exactly once and no larger product is examined. For a tuple $(p_1, \dots, p_m)$ the divisor list is generated by starting from $\{1\}$ and, for each prime p_i, adjoining p_i times every divisor already present; squarefreeness makes this exactly the 2^m divisors.

The program then tests the maximal proper divisor sets $\mathcal{D}(n/p)$. By lemma 2.3, a candidate can be discarded as soon as one of these is non-Sidon. For every surviving n, all

$$\binom{2^{\omega(n)}+1}{2}$$

unordered pairs of divisors are formed, their sums are sorted, and the largest run length is read off as $R(n)$ using exact 64-bit integer arithmetic. The source contains assertions for every row count below, for the number of surviving sharp cases, and for the complete factorization, multiplicity, and repeated sum of each of the two survivors.

$\omega(n)$	Products checked	All $\mathcal{D}(n/p)$ Sidon
5	3142839	302479
6	479316	0
7	28998	0
8	347	0

In total, 3651500 squarefree products are checked. Exactly two candidates survive with $R(n) \geq 3$, namely n_1 and n_2 from theorem 1.4.

A second C++17 implementation repeats the census with different internal representations and returns the same row counts, the same maximal-proper-Sidon counts, and exactly the same two survivors with repeated sums 2770 and 2802. Thus theorem 1.4 is reproduced by two independent exhaustive searches. Their sources, transcripts, and checksums are included in the verification package described in section 13.

There are two small points needed to turn this finite computation into a proof that they are globally the first two examples. First, theorem 1.1 excludes $\omega(n) \leq 4$ altogether. Second,

$$2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 = 223092870 > 82643495,$$

so no integer below the second example can have nine distinct prime factors. Thus the search covers every possible squarefree minimal obstruction below n_2.

9 The first cyclic template

We now force the cyclic lemma to produce the pattern seen in the smallest example $7 \cdot 11 \cdot 31 \cdot 89 \cdot 383$. The target identity is

$$b + cd = d + ae = e + abc.$$

To match it with lemma 7.1, we take the three free coordinates to be b, d, e and choose the multipliers so that the remaining factors are a and c.

Set

$$(x_1, x_2, x_3) = (b, d, e), \qquad (r_1, r_2, r_3) = (ac, c, a).$$

Then lemma 7.1 becomes

$$b + cd = d + ae = e + abc.$$

The primitive vector before division is

$$A = c(a - 1) + 1,$$

$$B = a^2c - a + 1,$$

$$C = ac^2 - ac + 1.$$

For distinct primes b, d, e, the common multiplier in lemma 7.1 is 1. Let $g = \gcd(A, B, C)$. Since

$$B - (a + 1)A = c - 2a,$$

we have $c \equiv 2a \pmod{g}$, and substituting into A gives

$$g \mid 2a^2 - 2a + 1.$$

Write

$$2a^2 - 2a + 1 = gh, \qquad c = gt + 2a.$$

Then

$$b = (a - 1)t + h,$$

$$d = a^2t + (a + 1)h,$$

$$e = agt^2 + a(4a - 1)t + (2a + 1)h.$$

Conversely, these formulas satisfy

$$b + cd = d + ae = e + abc$$

identically whenever $gh = 2a^2 - 2a + 1$. The common sum is

$$a^2gt^2 + 4a^3t + (2a^2 + 2a + 1)h.$$

Proposition 9.1 (Structure of Template I)**.** *The quadratic polynomial $e(t)$ has discriminant*

$$a(a - 4).$$

The pairwise resultants of b, c, d, e with respect to t are

$$-1,\ -h,\ h,\ 1 - a,\ g,\ ah.$$

If $g = 1$, then $1 + ac = b + c$; if $h = 1$, then $b + c = ab + d$. Thus a divisibility-minimal prime solution requires $g, h > 1$.

Interchanging g and h and replacing t by gt/h fixes c and scales b, d, e and the common sum by g/h.

Proof. All assertions follow by direct expansion and elimination. The resultant identities and the duality are also verified independently in the exact symbolic supplement. $\square$

For prime $a = 2, 3, 5$, the auxiliary values $5, 13, 41$ are prime. The first prime a giving a nontrivial factorization is

$$a = 7, \qquad 2a^2 - 2a + 1 = 85 = 5 \cdot 17.$$

This is the first point at which the auxiliary integer has a nontrivial factorization. Its two factors can be assigned in either order, giving the two canonical branches below.

9.1 The branch $(g,h)=(5,17)$

Set

$$\begin{aligned} p_1 &= 7,\\ p_2 &= 5t+14,\\ p_3 &= 6t+17,\\ p_4 &= 49t+136,\\ p_5 &= 35t^2+189t+255. \end{aligned}$$

Lemma 9.2 (Exact family-audit criterion)**.** *For each of the four branches below, form the* 32 *squarefree divisor polynomials and the* 528 *unordered pair-sum polynomials. Group identical pair-sum polynomials symbolically. Suppose that*

- *(a) there is exactly one polynomial identity class, namely the prescribed class of three representations;*
- *(b) every two representations in that identity class jointly use all five prime factors;*
- *(c) every positive integer root of the difference of two nonidentical pair-sum polynomials violates the branch's prime hypotheses, except for any parameter explicitly excluded in the theorem statement.*

Then the branch is divisibility-minimal non-Sidon, its prescribed triple sum is unique, and $R(n)=3$*.*

Proof. At a fixed parameter for which the displayed factor polynomials are distinct primes, every divisor is represented by exactly one subset-product polynomial, so the 528 unordered pairs exhaust all divisor-sum representations. Condition (a) identifies the only collision valid identically in the parameter. Every other collision must occur at a positive integer root of a nonzero difference polynomial and is therefore excluded by (c).

It remains to justify divisibility-minimality. Suppose that a proper divisor $m \mid n$ had a non-Sidon divisor set. The two equal-sum representations in $\mathcal{D}(m)$ are also two representations in $\mathcal{D}(n)$. There are only two possibilities. If their two pair-sum polynomials are nonidentical, then their parameter is a positive integer root of a nonzero difference polynomial, contradicting (c). If their pair-sum polynomials are identical, then by (a) both representations belong to the unique prescribed identity class. But condition (b) says that any two representations in that class jointly use all five prime factors of n; consequently no proper divisor $m<n$ can contain all four summands of both representations. This is again a contradiction. Hence every proper divisor set is Sidon, and the branch is divisibility-minimal. Conditions (a) and (c) also show that, at an admissible parameter, the full divisor set has no collision other than the prescribed identity class. That class has exactly three representations, so the triple sum is unique and $R(n)=3$. □

An exact symbolic audit verifies the three conditions of lemma 9.2 for all four branches. For a nonzero difference polynomial $H(T)$, every positive integer root divides the nonzero constant coefficient after any power of T is removed, so all possible exceptional parameters can be checked by exact integer arithmetic. The four branches yield precisely the exceptional sets

$$\{3,7\}, \qquad \{3\}, \qquad \{3\}, \qquad \{1\}.$$

A pure-Python implementation independent of the Maple audits reproduces the same result. The source files, exact counts, and transcripts are listed in section 13.

Theorem 9.3 (First $a = 7$ branch)**.** *Let t be a positive integer such that p_2, p_3, p_4, p_5 are prime. Then*

$$n = p_1 p_2 p_3 p_4 p_5$$

is divisibility-minimal non-Sidon, $R(n) = 3$, and the unique sum of multiplicity three is

$$245t^2 + 1372t + 1921.$$

Proof. The template identity gives

$$p_3 + p_2 p_4 = p_4 + 7p_5 = p_5 + 7p_2 p_3 = 245t^2 + 1372t + 1921.$$

For this branch, the audit criterion leaves only the additional positive collision parameters $t = 3$ and $t = 7$. Both violate the prime hypotheses: $6t + 17 = 35$ at $t = 3$, while $5t + 14 = 49$ at $t = 7$. Hence lemma 9.2 gives divisibility-minimality, uniqueness of the triple sum, and $R(n) = 3$. ☐

9.2 The branch $(g, h) = (17, 5)$

Set

$$\begin{aligned} q_1 &= 7, \\ q_2 &= 17t + 14, \\ q_3 &= 6t + 5, \\ q_4 &= 49t + 40, \\ q_5 &= 119t^2 + 189t + 75. \end{aligned}$$

Theorem 9.4 (Second $a = 7$ branch)**.** *Let t be a positive integer such that q_2, q_3, q_4, q_5 are prime. Then*

$$n = q_1 q_2 q_3 q_4 q_5$$

is divisibility-minimal non-Sidon, $R(n) = 3$, and the unique sum of multiplicity three is

$$833t^2 + 1372t + 565.$$

Proof. The identity is

$$q_3 + q_2 q_4 = q_4 + 7q_5 = q_5 + 7q_2 q_3 = 833t^2 + 1372t + 565.$$

For this branch, the only further positive collision parameter is $t = 3$, where $q_2 = 65$ is composite. The conclusion therefore follows from lemma 9.2. ☐

At $t = 1$, theorem 9.4 gives

$$(q_1, q_3, q_2, q_4, q_5) = (7, 11, 31, 89, 383),$$

which is the smallest example in theorem 1.4.

10 The second cyclic template

The second-smallest example has a different-looking triple collision. It is obtained from the same cyclic lemma by changing which prime product is used as the first coordinate. We now target

$$ab + cd = abc + e = d + ae.$$

Set

$$(x_1, x_2, x_3) = (ab, d, e), \qquad (r_1, r_2, r_3) = (c, c, a).$$

The cyclic identity becomes

$$ab + cd = abc + e = d + ae.$$

The three primitive coordinates before division are

$$A = c(a - 1) + 1, \qquad B = ac - a + 1, \qquad C = c^2 - c + 1.$$

Let $h = \gcd(A, B, C)$. Since $B - A = c - a$, one obtains

$$h \mid a^2 - a + 1.$$

Write

$$a^2 - a + 1 = hk.$$

The condition $A/h = ab$ yields a parameter u and the formulas

$$b = (a - 1)u + k,$$

$$c = ahu + a^2 + 1,$$

$$d = a^2u + (a + 1)k,$$

$$e = a^2hu^2 + a(2a^2 + 1)u + (a^2 + a + 1)k.$$

They satisfy the template identity, whose common sum is

$$a^3hu^2 + 2a^2(a^2 + 1)u + (a^3 + a^2 + 2a + 1)k.$$

Proposition 10.1 (Structure of Template II). *The quadratic polynomial $e(u)$ has discriminant*

$$-3a^2.$$

If $h = 1$, then

$$1 + ac = a + d = ab + c;$$

if $k = 1$, then

$$1 + ad = ab + c.$$

Thus divisibility-minimality requires $h, k > 1$.

Proof. Direct expansion gives the identity and common sum. The discriminant reduction and boundary identities are verified symbolically in the supplement. □

The first prime a for which $a^2 - a + 1$ is composite is

$$a = 5, \qquad a^2 - a + 1 = 21 = 3 \cdot 7.$$

Again the first useful auxiliary factorization has two nontrivial factors, and interchanging them produces two natural branches.

10.1 The branch $(h, k) = (3, 7)$

Set

$$\begin{aligned} r_1 &= 5, \\ r_2 &= 4u + 7, \\ r_3 &= 15u + 26, \\ r_4 &= 25u + 42, \\ r_5 &= 75u^2 + 255u + 217. \end{aligned}$$

Theorem 10.2 (First $a = 5$ branch)**.** *Let u be a positive integer such that r_2, r_3, r_4, r_5 are prime. Then*

$$n = r_1 r_2 r_3 r_4 r_5$$

is divisibility-minimal non-Sidon, $R(n) = 3$, and the unique sum of multiplicity three is

$$375u^2 + 1300u + 1127.$$

Proof. The three representations are

$$5r_2 + r_3 r_4 = 5r_2 r_3 + r_5 = r_4 + 5r_5.$$

For this branch, the only additional positive collision parameter is $u = 3$, where $r_4 = 117$ is composite. The conclusion follows from lemma 9.2. ☐

At $u = 1$, this gives

$$(5, 11, 41, 67, 547),$$

the second-smallest example in theorem 1.4.

10.2 The branch $(h, k) = (7, 3)$

Set

$$\begin{aligned} s_1 &= 5, \\ s_2 &= 4u + 3, \\ s_3 &= 35u + 26, \\ s_4 &= 25u + 18, \\ s_5 &= 175u^2 + 255u + 93. \end{aligned}$$

Theorem 10.3 (Second $a = 5$ branch)**.** *Let u be a positive integer different from 1 such that s_2, s_3, s_4, s_5 are prime. Then*

$$n = s_1 s_2 s_3 s_4 s_5$$

is divisibility-minimal non-Sidon, $R(n) = 3$, and the unique sum of multiplicity three is

$$875u^2 + 1300u + 483.$$

Proof. The cyclic identity gives the stated triple sum. Here $u = 1$ is the only positive nonidentity collision parameter not already excluded by compositeness, and hence the only simultaneous-prime exceptional parameter. Indeed, at $u = 1$,

$$(s_1, s_2, s_3, s_4, s_5) = (5, 7, 61, 43, 523)$$

and

$$1 + 5 \cdot 61 = 5 + 7 \cdot 43 = 306.$$

For every other simultaneous-prime parameter, the full divisor set has exactly the prescribed triple identity, and the full-support condition in lemma 9.2 implies that every proper divisor set is Sidon. □

11 Conditional infinitude

The four preceding families are unconditional implications: whenever the displayed polynomial values are prime, the resulting integer has the stated Sidon property. To conclude that each branch supplies infinitely many examples, one needs a simultaneous-prime-values conjecture.

We use the standard form of Schinzel's Hypothesis H: a finite family of irreducible integer polynomials with positive leading coefficients and no fixed prime divisor assumes simultaneous prime values infinitely often [4].

Theorem 11.1 (Conditional infinitude)**.** *Assume Schinzel's Hypothesis H. Each of the four canonical branches in theorems 9.3, 9.4, 10.2 and 10.3 contains infinitely many divisibility-minimal non-Sidon integers with five prime factors and $R(n) = 3$.*

Proof. The three linear polynomials in every branch are primitive and irreducible. The four quadratic coefficient triples are

$$(35, 189, 255), \quad (119, 189, 75), \quad (75, 255, 217), \quad (175, 255, 93),$$

and each has coefficient gcd 1, so the quadratics are primitive as well. Their discriminants are 21 in the two $a = 7$ branches and -75 in the two $a = 5$ branches; hence they are irreducible over $\mathbb{Z}$.

For a prime $p > 5$, let the four branch polynomials be $f_1, \dots, f_4$ and put

$$P(T) = \prod_{i=1}^{4} f_i(T) \in \mathbf{F}_p[T].$$

Primitivity ensures that no f_i becomes the zero polynomial modulo p. Hence P is a nonzero polynomial of degree $5 < p$, so it has at most five roots in $\mathbf{F}_p$ and cannot vanish on every residue class. This checks the tuple as a whole, which is the admissibility condition. For $p = 2, 3, 5$, Table 4 gives an explicit residue at which all four polynomials are nonzero.

Table 4: Free residue classes proving admissibility at $p = 2, 3, 5$.

Branch	modulo 2	modulo 3	modulo 5
I.1 $(5, 17)$	1	1	2
I.2 $(17, 5)$	1	1	1
II.1 $(3, 7)$	1	1	0
II.2 $(7, 3)$	1	1	0

Hypothesis H therefore gives infinitely many simultaneous prime-value parameters in each branch. In Branch II.2, deleting the single exceptional parameter $u = 1$ leaves infinitely many parameters covered by theorem 10.3. □

12 Consequences for the four OEIS sequences

The notation introduced in the introduction makes the relation between the four sequences immediate:

$$\mathrm{A398434}(n) = \alpha(n), \qquad \mathrm{A398435}(n) = C(n), \qquad \mathrm{A398577}(n) = R(n),$$

while A398433 records the divisibility-minimal indices at which these quantities first detect a failure of the Sidon property. For every positive integer n,

$$\mathrm{A398577}(n) = 1 \iff \mathrm{A398435}(n) = 0 \iff \mathrm{A398434}(n) = |\mathcal{D}(n)|.$$

Thus A398433 consists precisely of the divisibility-minimal indices for which any, and hence all, of these equivalent Sidon equalities fail.

The threshold theorems can now be read directly as information about A398577 on the squarefree terms of A398433: with at most four distinct prime factors its value is at most 2, and with at most five distinct prime factors its value is at most 3. Both bounds are attained. Moreover, theorem 1.4 identifies the first two squarefree terms of A398433 at which A398577 takes the value 3.

The four parametric families give a stronger simultaneous statement. They do not merely produce terms of A398433; because each admissible divisor set has exactly one collision and that collision has three representations, the other three sequences are determined at the same time.

Corollary 12.1 (Simultaneous values of the four OEIS sequences)**.** *Every integer n produced by one of the four canonical branches in theorems 9.3, 9.4, 10.2 and 10.3 is a term of A398433 and satisfies*

$$\mathit{A398434}(n) = 30, \qquad \mathit{A398435}(n) = 2, \qquad \mathit{A398577}(n) = 3.$$

Proof. Each such n is squarefree with five prime factors, so $|\mathcal{D}(n)| = 32$. The family audit shows that there is exactly one repeated sum and that it has three representations, involving six distinct divisors. Hence $R(n) = 3$ and its contribution to the collision excess is $3 - 1 = 2$, so $C(n) = 2$.

A Sidon subset must destroy all but one of the three representations. Since distinct representations of the same sum cannot share a divisor, one deletion can destroy at most one of them; at least two deletions are therefore necessary. Conversely, deleting one divisor from each of two representing pairs leaves only one representation of the repeated sum and creates no new collision. Hence $\alpha(n) = 32 - 2 = 30$. Divisibility-minimality was proved in the corresponding family theorem. ☐

13 Computational certification and reproducibility

Several theorems above contain a finite computer-assisted step. This section gathers the implementation and reproduction details that were deliberately kept out of the main narrative. The mathematical domains of the threshold audits are fixed by lemmas 3.4 and 4.1 to 4.3; the programs enumerate those finite domains and verify exact algebraic certificates. No floating-point arithmetic is used in a proof, and the threshold theorems do not depend on testing primes only up to a chosen bound.

Maple is used for symbolic expansion, coefficient-sign tests, exact factorization, modular certificates, and comparisons of pair-sum polynomials. A separate C++17 program performs the bounded search used only in theorem 1.4. The main accounting is summarized below.

Result	Exact outcome
Threefold threshold	$6092 \to 6056 + 36$; the 36 residual orientations form 16 equation groups under the safe grouping of lemma 4.6, all closed in Appendix A.
Odd five-prime fourfold threshold	$296960 \to 509 \to 498 + 11 \to 0$; the 498 are explicit modular-certificate records and the 11 residuals are closed in section 6.
Small/even fourfold threshold	$4384 \to 0$ and $61440 \to 38 \to 37 + 1 \to 0$; the 37 are independent modular certificates and the final case has the printed elementary closure.
First $a = 7$ branch	32 divisor polynomials and 528 pair sums; one identity class of size 3; the only other positive collision parameters are $t = 3, 7$, both incompatible with the prime hypotheses.
Second $a = 7$ branch	The same full and omission audits; only $t = 3$ is an additional positive collision parameter, with a composite required factor.
Two $a = 5$ branches	Both symbolic audits pass; Branch II.2 has the unique simultaneous-prime nonminimal exception $u = 1$.
Two smallest examples	3651500 squarefree products checked; exactly 81365669 and 82643495 survive with $R(n) \geq 3$.

What the threshold certificate files contain

The threefold program generates every labeled block assignment and every unordered factorization exactly as in lemma 4.2. It asserts each row count in table 1, the total 6092, the sign-closed total 6056, the residual total 36, the existence of exactly 16 canonical equation keys, and the multiplicity of every key.

The odd five-prime stage-one program uses the canonical fourfold block convention of lemma 4.3. For each of the six types it asserts both the orientation count and the sign-stage residual count in table 3; it also asserts the totals 296960 and 509. The second stage constructs for each residual a record

$$[\text{case index}, (c_1, c_2, c_3), p, \text{factor}(g)],$$

with g as in lemma 4.4. The complete v7 residual ledger and certificate ledger expose these records directly: there are exactly 498 stored modular certificates, and their complementary index set inside $\{1, \ldots, 509\}$ is exactly the eleven indices printed in section 6. The stage-three file checks the oriented sums at those indices and the polynomial identities used in their elementary closures. The manuscript itself supplies the final inequalities and divisibility arguments, so no opaque “exception accepted” instruction is used as a proof.

The historical small/even Maple file reports 38 modular closures. The proof in this version does not rely on that uniform count: the independent v8 Python re-audit reconstructs the same 38 residual systems, verifies 37 modular certificates, and closes residual 31 by the elementary argument in section 6.1. Thus the independently checked accounting is $61440 \to 38 \to 37 + 1 \to 0$. This v8 verification and its separate stored-record checker are retained unchanged in the present package.

The four parametric families likewise retain the independent pure-Python checker `family_symbolic_checker_v8.py`, in addition to the original Maple audits. It enumerates all 32 divisor polynomials and 528 unordered pair-sum polynomials for each branch, verifies the unique identity class and its pairwise full support, and determines all positive integer roots of nonidentity difference polynomials by exact integer arithmetic. Thus the family conclusions do not depend on the Maple implementation alone.

For the threefold threshold, `threefold_independent_verifier_v9.py` independently reconstructs the complete $6092 \to 6056 + 36$ sign audit, while `verify_threefold_residual_records_v9.py` checks the stored 36 residual records without re-running the orientation enumeration. For additional transparency, `orientation_count_ledger_v7.py` independently evaluates the elementary formulas behind all orientation counts in tables 1 to 3.

How to reproduce the calculations

Every main Maple program is self-checking: an unexpected count, identity, factorization, exception list, or coefficient test raises an error before the final `PASSED` line. The recorded runs were made in Maple 2026. With the current directory set to `anc`, execute the following files in separate fresh sessions:

```
read "threefold_four_prime_obstruction_audit_final_corrected.mpl";
read "START_HIER_NUR_TEIL_2_KLEIN_GERADE_FINAL.mpl";
read "parametric_five_prime_sharpness_symbolic_audit_fast_corrected.mpl";
read "parametric_five_prime_second_branch_full_audit.mpl";
read "general_triple_collision_structure_audit_corrected.mpl";
read "second_triple_collision_structure_audit_corrected.mpl";
read "second_triple_collision_two_branch_symbolic_audit.mpl";
```

For the odd five-prime threshold, change to `anc/odd_five_prime_strict_package` and run

```
read "RUN_1_ODD_FIVE_PRIME_STRICT.mpl";
```

The exhaustive search for theorem 1.4 can be reproduced in two independent implementations from the `anc` directory:

```
g++ -O3 -std=c++17 two_smallest_sharp_examples_search.cpp -o search
./search
g++ -O3 -std=c++17 \
  two_smallest_sharp_examples_search_independent_v9.cpp -o search_v9
./search_v9
```

The independent exact verification scripts are run with Python 3:

```
python3 orientation_count_ledger_v7.py
python3 threefold_independent_verifier_v9.py
python3 verify_threefold_residual_records_v9.py
python3 even_five_independent_verifier_v8.py
python3 verify_even_five_certificate_records_v8.py
python3 family_symbolic_checker_v8.py
```

The package contains the final console transcripts and a SHA-256 manifest for every ancillary source, proof ledger, and stored output file. The manifest authenticates source code rather than a compiler-dependent executable. Appendix B gives the file-by-file inventory.

AI-assisted research disclosure

The mathematical exploration and preparation of this manuscript were carried out with substantial assistance from OpenAI's ChatGPT. The system was used interactively to explore proof strategies, develop and revise arguments, design exact symbolic computations, identify possible gaps, and edit the manuscript. The author selected the research questions and final statements, executed the stated Maple and C++ computations, reviewed their outputs, and remains responsible for all mathematical claims.

14 Open problems

The threshold $R(n) \geq 4 \Rightarrow \omega(n) \geq 6$ leaves the six-prime case as the next natural boundary.

Question 14.1. Does every squarefree divisibility-minimal non-Sidon integer with $R(n) \geq 4$ satisfy $\omega(n) \geq 7$?

The present paper does not claim this stronger bound. A substantial six-prime case analysis exists, but a complete order-preserving closure has not been established, so it is deliberately kept outside the theorem set.

The second question is structural rather than quantitative.

Question 14.2. Do the two cyclic templates classify all five-prime divisibility-minimal examples with $R(n) = 3$?

The exhaustive search shows that the two smallest examples arise from the two different templates, but this does not amount to a global classification of all five-prime triple collisions.

Question 14.3. For every nontrivial factorization branch in either template, does simultaneous primality imply divisibility-minimality, apart from finitely many explicitly describable exceptions?

The four canonical branches support this expectation. The general statement, however, would require control of all auxiliary factorization branches rather than the four treated here.

A The sixteen residual threefold groups

Let $a < b < c < d$ be distinct primes. After the uniform sign stage, the 36 residual orientations give the following 16 equation groups. We include the elementary closures here so that the last step of the threefold threshold can be checked without consulting the code.

1 (six orientations). $a + bc = ac + d = ad + b$. Put $q = d - c > 0$. From $ac + d = ad + b$,

$$(a-1)q = c - b.$$

From $a + bc = ac + d$,

$$q = a + c(b - a - 1).$$

If $b \geq a + 2$, then $q > c$, contradicting $(a-1)q = c - b < c$. Hence $b = a + 1$, so $a = 2, b = 3$. Then $q = 2, c = 5, d = 7$. The corresponding $n = 210$ is not minimal because $\mathcal{D}(6)$ is non-Sidon.

2 (two orientations). $a + bc = ab + d = c + d$**.** The last equality gives $ab = c$, impossible.

3 (two orientations). $ab + d = ac + b = c + d$**.** The first and last expressions give $ab = c$, impossible.

4 (two orientations). $1 + abc = ab + d = c + d$**.** The last two expressions give $ab = c$, impossible.

5 (four orientations). $1 + abc = a + bd = ad + bc$**.** Reducing the first equality modulo b gives $b \mid a - 1$, impossible.

6 (two orientations). $1 + abc = a + cd = ac + bd$**.** Reducing the first equality modulo c gives $c \mid a - 1$, impossible.

7 (two orientations). $1 + abc = a + cd = ad + bc$**.** Again reduction modulo c gives $c \mid a - 1$, impossible.

8 (four orientations). $1 + abd = a + cd = abc + d$**.** Reduction modulo d gives $d \mid a - 1$, impossible.

9 (two orientations). $1 + abc = ac + bd = b + cd$**.** From the last two expressions,

$$b(d - 1) = c(d - a).$$

Thus $c \mid d - 1$; write $d = 1 + cq$. Substitution gives

$$c\big(a(b - 1) - bq\big) = b - 1,$$

impossible because $0 < b - 1 < c$.

10 (two orientations). $1 + abc = ad + bc = b + cd$**.** The first and last expressions give

$$c(ab - d) = b - 1.$$

The left side is nonpositive or at least c, whereas $0 < b - 1 < c$.

11 (two orientations). $1 + abc = ad + bc = bd + c$**.** The first and last expressions give

$$\frac{d}{c} = a - \frac{1}{b} + \frac{1}{bc},$$

while the last two give

$$\frac{d}{c} = \frac{b - 1}{b - a}.$$

If $b = a + 1$, then $a = 2, b = 3$ and the two ratios lie on opposite sides of 2. If $b \geq a + 2$ and $a \geq 3$, then

$$\frac{b - 1}{b - a} \leq \frac{a + 1}{2} < a - \frac{1}{3}.$$

If $a = 2, b \geq 5$, the second ratio is at most $4/3$ and the first exceeds $2 - 1/5$.

12 (two orientations). $1 + abd = abc + d = b + cd$**.** The first equality implies

$$(ab - 1)d = abc - 1 = (ab - 1)c + (c - 1),$$

so $ab - 1 \mid c - 1$. Write

$$c = 1 + r(ab - 1), \qquad d = c + r.$$

Then $c < d < 2c$. Modulo c, the second equality gives $d \equiv b \pmod{c}$, hence $d = c + b$ and $r = b$. Therefore

$$c = ab^2 - b + 1, \qquad d = ab^2 + 1.$$

If a is odd, d is even and greater than 2; if $a = 2$, c is even and greater than 2.

13 (one orientation). $1 + abc = 1 + bd = ad + bc$**.** The first equality gives $ac = d$, impossible.

14 (one orientation). $1 + abc = 1 + cd = ac + bd$**.** The first equality gives $ab = d$, impossible.

15 (one orientation). $1 + abc = 1 + cd = ad + bc$**.** Again $ab = d$, impossible.

16 (one orientation). $1 + abd = 1 + cd = abc + d$**.** The first equality gives $ab = c$, impossible.

B Verification source inventory

The companion verification package distributed with this OEIS supporting note contains the following exact files in the `anc/` directory. The same inventory appears in `anc/README.txt`.

- `threefold_four_prime_obstruction_audit_final_corrected.mpl`: complete 6092-orientation threefold threshold audit.
- `threefold_four_prime_residual_proof.txt`: human-readable closure of the 16 residual groups.
- `threefold_independent_verifier_v9.py`: independent exact Python/SymPy reconstruction of all 6092 threefold orientations, the 6056 strict sign closures, and the 36 residuals.
- `threefold_independent_verifier_v9.txt`: stored successful output of that independent reconstruction.
- `THREEFOLD_RESIDUALS_36_v9.json` and `THREEFOLD_RESIDUALS_36_v9.txt`: complete machine-readable and compact text ledgers of the 36 threefold residual orientations.
- `verify_threefold_residual_records_v9.py`: non-generating verifier for the stored 36 residual records and their 16 canonical key multiplicities.
- `verify_threefold_residual_records_v9.txt`: stored successful output of the residual-record verifier.
- `odd_five_prime_strict_package/`: odd five-prime strict sign, modular, and exceptional audit, started with `RUN_1_ODD_FIVE_PRIME_STRICT.mpl`.
- `START_HIER_NUR_TEIL_2_KLEIN_GERADE_FINAL.mpl`: small and even strict fourfold audit.
- `parametric_five_prime_sharpness_symbolic_audit_fast_corrected.mpl`: first $a = 7$ branch.

- parametric_five_prime_second_branch_full_audit.mpl: second $a = 7$ branch.
- general_triple_collision_structure_audit_corrected.mpl: structure, resultants, discriminant, and duality for Template I.
- second_triple_collision_structure_audit_corrected.mpl: structure and discriminant for Template II.
- second_triple_collision_two_branch_symbolic_audit.mpl: complete symbolic audits of the two $a = 5$ branches.
- two_smallest_sharp_examples_search.cpp: exhaustive C++17 search proving the two smallest sharp examples.
- two_smallest_sharp_examples_search_output.txt: stored output of that search.
- two_smallest_sharp_examples_search_independent_v9.cpp: second exact C++17 implementation of the complete smallest-example census, using different divisor and pair-sum data structures.
- two_smallest_sharp_examples_search_independent_v9_output.txt: stored successful output of the independent v9 search.
- orientation_count_ledger_v7.py: independent standard-library check of every combinatorial orientation count printed in the manuscript.
- orientation_count_ledger_v7.txt: stored output of the independent orientation-count check.
- COMPUTATIONAL_CERTIFICATE_SPEC_v7.txt: plain-text description of the enumeration conventions and exact sign/modular certificate format.
- EXCEPTIONAL_CLOSURES_v7.txt: the eleven odd five-prime residual closures in a compact plain-text proof ledger.
- ODD_FIVE_RESIDUALS_509_v7.txt: the complete indexed list of all 509 odd five-prime sign-stage residual systems.
- ODD_FIVE_MODULAR_CERTIFICATES_v7.json: complete machine-readable records for the 498 uniform modular certificates.
- ODD_FIVE_MODULAR_CERTIFICATES_v7.tsv: compact human-readable table of the same 498 certificate records.
- verify_odd_five_modular_certificates_v7.py: independent exact verifier for the stored odd five-prime certificate records; it does not search for certificates.
- verify_odd_five_modular_certificates_v7.txt: stored successful output of that verifier.
- even_five_independent_verifier_v8.py: independent exact reconstruction of all 61440 even five-prime orientations, the 38 strict-sign residuals, 37 modular certificates, and the single elementary residual.
- even_five_independent_verifier_v8.txt: stored successful output of the even-five verifier.
- EVEN_FIVE_RESIDUALS_38_v8.txt: complete indexed list of the 38 even five-prime sign-stage residual systems.
- EVEN_FIVE_CERTIFICATES_v8.json: machine-readable records for the 37 independently verified even-five modular certificates and the index of the elementary closure.
- verify_even_five_certificate_records_v8.py: independent record verifier that reads the stored 37 certificates without searching for new ones.
- verify_even_five_certificate_records_v8.txt: stored successful output of that record verifier.
- EVEN_FIVE_v8_NOTE.txt: transparent note on the historical uniform-38 Maple classification and the v8 independent $37 + 1$ proof accounting.
- family_symbolic_checker_v8.py: independent pure-Python exact verification of the four parametric family audits using the integer-root theorem.

- `FAMILY_INDEPENDENT_VERIFIER_v8.txt`: stored successful output of the independent family verifier.
- `FAMILY_AUDIT_LEDGER_v7.txt`: branch-by-branch Maple ledger retained as a second implementation of the four parametric family audits.
- `console_outputs/`: recorded final Maple console transcripts used in the theorem ledger.
- `SHA256SUMS.txt`: SHA-256 checksums for the complete ancillary package.

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